STAT 709: Homework 8

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All three problems are required. Complete every part of each problem.

1. Fisher information and sufficient statistics

Jun Shao, *Mathematical Statistics*, second edition, Chapter 3: Exercise 36, parts (a) and (b).

2. Bias–variance tradeoff and the Cramér–Rao bound

Let the sample X have density f_θ with respect to a fixed measure, where the scalar parameter θ ranges over an open interval. Let $T = T(X)$ be a statistic estimating θ, and write

$$b_T(\theta) = \mathbb{E}_\theta T - \theta, \qquad I(\theta) = \mathbb{E}_\theta\left[\left\{\frac{\partial}{\partial\theta}\log f_\theta(X)\right\}^2\right].$$

Assume the usual Cramér–Rao regularity conditions: f_θ is differentiable in θ, its support does not depend on θ, and differentiation may pass under the integrals of f_θ and Tf_θ. Also assume $0 < I(\theta) < \infty$ and $\mathbb{E}_\theta T^2 < \infty$. Prove that

$$\mathbb{E}_\theta[(T-\theta)^2] \geq \frac{[1 + b_T'(\theta)]^2}{I(\theta)} + b_T(\theta)^2.$$

Remark: A suitable amount of bias can reduce variance and lead to a smaller mean squared error. The next exercise gives an example.

3. Variance estimation

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$, where $n \geq 2$, $\mu \in \mathbb{R}$, and $\sigma^2 > 0$ are unknown. Write $\overline{X} = n^{-1}\sum_{i=1}^n X_i$ and define

$$S_n^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2, \qquad V_n = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2 = \frac{n-1}{n}S_n^2.$$

(i) Derive the bias and variance of V_n as an estimator of σ^2. Show that its mean squared error is smaller than that of S_n^2.

(ii) Among estimators $c_n S_n^2$, where the nonrandom multiplier c_n may depend on n but not on the data or unknown parameters, find the value that minimizes mean squared error.

(iii) Derive the Fisher information matrix for $\theta = (\mu, \sigma^2)^\top$ based on all n observations. The sample mean $\overline{X}$ and sample variance S_n^2 are the uniformly minimum-variance unbiased estimators (UMVUEs) of μ and σ^2, respectively. Compare the covariance matrix of $\widehat{\theta} = (\overline{X}, S_n^2)^\top$ with the matrix Cramér–Rao bound. State whether each component attains its bound.